

SmartCare Hospital AI Decision Support System

Coursework Technical Report (Option C: Disease Risk Classification)
Module: CCS3440 – Artificial Intelligence | SLTC Research University

1. Problem Definition & Objectives

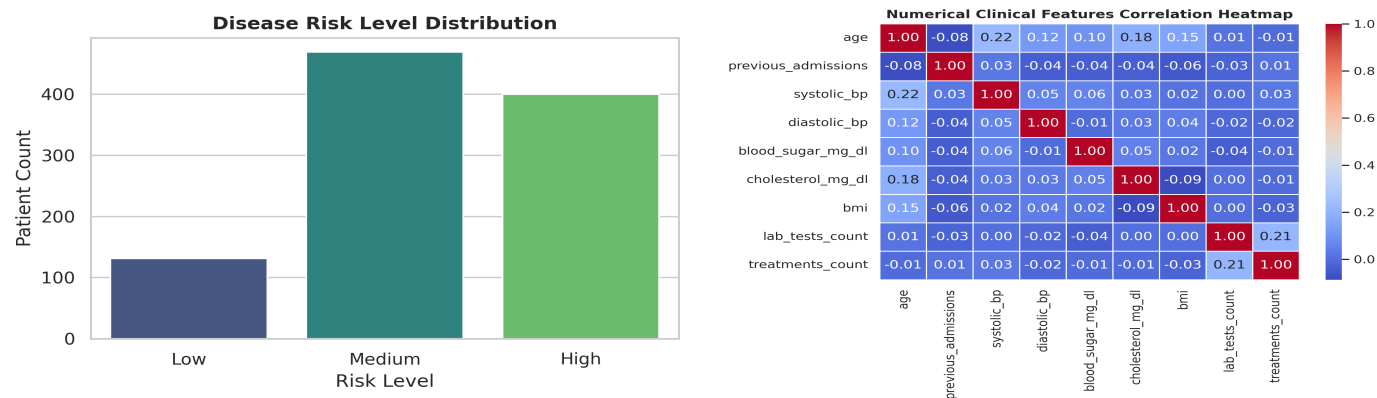
Early stratification of patient disease risk enables healthcare providers to prioritize preventative clinical interventions, optimize bed allocation, and reduce hospital readmissions. Option C formulates this challenge as a **multi-class classification problem** to categorize patient risk into three distinct tiers: **Low**, **Medium**, and **High** risk levels. The dataset incorporates patient demographics, clinical vitals (blood pressure, BMI), metabolic biomarkers (blood sugar, cholesterol), and diagnostic history.

2. System Architecture & Pipeline Workflow

Step 1: Ingestion	Raw dataset loading (1,000 patient records) and data dictionary verification.
Step 2: Preprocessing	Clinical feature selection (13 variables), One-Hot Encoding, StandardScaler.
Step 3: Model Development	Candidate training: Logistic Regression, Random Forest, Decision Tree.
Step 4: Model Evaluation	Stratified test benchmark (Accuracy, Weighted Precision, Recall, F1-Score).
Step 5: Explainable AI	SHAP summary analysis to interpret feature contributions.
Step 6: Prototype UI	Streamlit clinical decision support app with physiological boundary checks.

3. Exploratory Data Analysis (EDA)

Exploratory analysis demonstrates marked class boundaries across key metabolic vitals. Blood sugar, cholesterol, and systolic blood pressure exhibit strong positive associations with elevated risk categories.



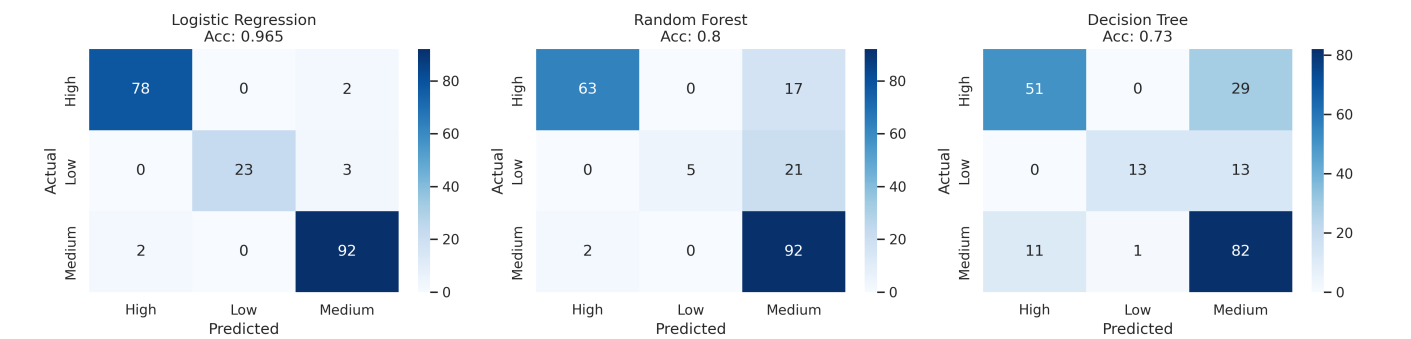
4. Feature Engineering & Preprocessing Workflow

To avoid data leakage, administrative billing amounts and post-admission operational targets were excluded. Categorical features were encoded with One-Hot encoding, and numerical attributes normalized via StandardScaler fitted strictly on the training set.

```
# Clinical Preprocessing Snippet
clinical_features = ['age', 'gender', 'blood_group', 'department', 'diagnosis',
                    'previous_admissions', 'systolic_bp', 'diastolic_bp',
                    'blood_sugar_mg_dl', 'cholesterol_mg_dl', 'bmi',
                    'lab_tests_count', 'treatments_count']
X_encoded = pd.get_dummies(df[clinical_features], columns=['gender', 'blood_group', 'department', 'diagnosis'], drop_first=True)
scaler = StandardScaler()
X_train_scaled[num_cols] = scaler.fit_transform(X_train[num_cols])
X_test_scaled[num_cols] = scaler.transform(X_test[num_cols])
```

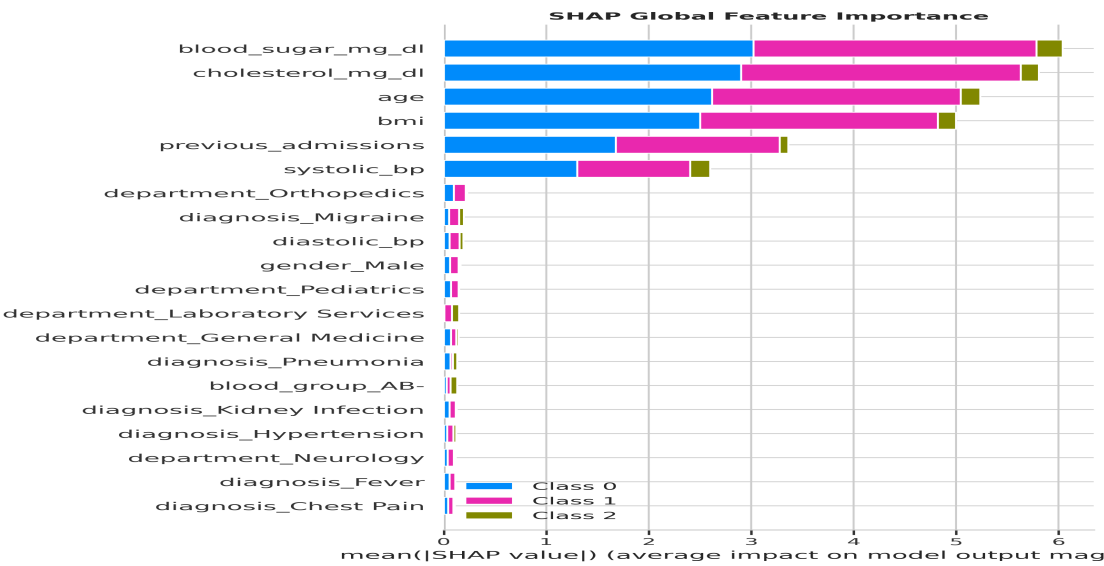
5. Model Benchmark & Evaluation Results

Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression (Best)	96.50%	0.9655	0.9650	0.9650
Random Forest (100 Trees)	80.50%	0.8112	0.8050	0.7984
Decision Tree (Depth=5)	73.00%	0.7320	0.7300	0.7291



6. Explainable AI Analysis (SHAP)

To provide clinical transparency, SHAP values were generated. Global summary plots confirm that elevated blood sugar, cholesterol, BMI, and systolic blood pressure are the primary contributors pushing classifications into the High Risk category.



7. Decision Support Prototype & Guardrails

The interactive Streamlit interface incorporates explicit clinical validation boundaries. Entering biologically implausible or extreme values triggers an immediate safety halt with clear user guidance rather than generating false out-of-distribution extrapolations.